Damian Kim

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EDUCATION

Duke University — **Expected Graduation: December 2026**

Bachelor of Science | Double Major: Computer Science and Mathematics — **GPA: 3.8**

Relevant Coursework: Theory & Algorithms of Machine Learning, Design & Analysis of Algorithms, Applied Stochastic Processes, Bayesian & Modern Statistics, Statistical Computing, Mathematical Finance, Algorithmic Trading, Computer Vision, Probability, Regression Analysis, Data Structures & Algorithms, Database Systems, Computer Systems, Discrete Mathematics

WORK EXPERIENCE

Mayo Clinic — **Rochester, MN**

IT Intern (AI FAST team) — *June 2023 - Aug 2024*

- Engineered scalable backend integrations of proprietary **ML APIs** with data processing pipelines using **Python** and **REST** frameworks, enabling efficient deployment and versioning of AI models in production environments
- Automated EDA by refactoring **Jupyter notebooks** into **Python scripts** for an automated integrated research pipeline
- Assisted with the team's migration from **Git** to **Azure DevOps**, redesigning source control and **CI/CD** workflows
- Created cloud-based data query pipelines on **GCP** using **BigQuery** and **Vertex AI**, replacing legacy on-prem **SQL**
- Implemented **frequency-based caching** and data clustering strategies for a 2+ TB radiology imaging dataset, significantly optimizing repeated query execution to reduce average scan volume by 20x and latency by 15-70%

Mayo Clinic — **Rochester, MN**

Data Science Mentee — *August 2021- May 2022*

- Improved patient prognosis time of major depressive disorder (MDD) in adolescents by creating a **decision support tool** based on a Markov Chain algorithm trained on 18 unique Hamilton Depression Rating Scale datasets
- Enhanced primary care experience by creating a real-time web interface with **HTML** coupled with **React, Node.js**, and **REST APIs** to provide practitioners with automated MDD prognosis prediction based on patient-input questionnaires

NASA Langley — **Hampton, VA**

Summer Research Mentee — *June - August 2021*

- Increased void analysis accuracy by creating and testing a **Keras Tensorflow**-based convolutional neural network (**CNN**) trained on cross-sectional images of polymer composites as compared to manual ImageJ analysis
- Identified failure modes and proposed AI architecture improvements for broader analysis applications in materials

PROJECTS

Golfie (Golf Simulation Software) — *April 2026 - Current*

- Hand-built a physics-based simulation modeling golf ball trajectory with numerical integration (Runge-Kutta)
- Created backend APIs for simulation state management, shot replay, and player stats with **Node.js, PostgreSQL, Redis**
- Deployed frontend interface using **Vercel**, establishing automated CI/CD workflows via GitHub Actions for updates
- Currently implementing real-time 3D rendering and state updates with WebGL, and collision mechanics with surfaces

Game Projects (rmhstudios.com) — *February - March 2026*

- Developed multiple real-time multiplayer video games with **Claude** Code, **React, and HTML5** Canvas
- Built backend services with **Node.js** for session management, matchmaking, and game state coordination via **REST APIs** and **WebSockets**. Built **PostgreSQL** schemas for user sessions, game state, and gameplay metrics, implementing indexing and query optimization (with Cursor) to support low latency reads/writes for concurrent gameplay w friends

Sate App — *June - July 2025*

- Built a full-stack **Flask** app with **MongoDB** database backend enabling restaurant recs from Tinder-esque swiping
- Tested a weighted group recommendation model incorporating negative user feedback for preference optimization, achieving ~65% higher consensus accuracy in survey testing compared to baseline random recommendation

SKILLS

Python (PyTorch, scikit-learn, TensorFlow), JS (Rest, React, Node), Java, SQL, Claude, XGBoost, Flask, GCP, Git, Azure